

Caleb Bellsmith

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TECHNICAL SKILLS

Languages: C++, HTML, L^AT_EX, Visual Basic

Software/Tools: SolidWorks, Fusion360, AutoCAD, VSCode, CLion, Bamboo, Arduino IDE

Fabrication: CNC Machining, 3D Printing, Soldering, Manual Prototyping

PROJECTS

Mechanical Exoskeleton

Personal Project

SolidWorks, Prototyping

2025 – Present

- Created a mechanical orthotic exoskeleton aimed at rehabilitating hand and wrist injuries.
- Optimized design for minimal weight and tool-less adjustability using SolidWorks modeling.

EMG-Activated Hand Exoskeleton

Independent Project

ESP32, C++, EMG Sensors

Dec 2025 – Present

- Developing a motorized orthotic using EMG sensors to detect neuromuscular signals for automated hand control.

Custom Cedar Strip Canoe

Personal Project

CAD, CNC Machining

2024 – Present

- Computer-modeled and fabricated a custom cedar strip canoe from scratch.
- Integrated CNC-machined internal molds with traditional woodworking to achieve high structural efficiency.

Adjustable Finger Prosthetic

Independent Project

User Testing, CAD, Iterative Design

2025 – Present

- Developed a functional finger prosthetic for a patient with scleroderma.
- Conducted repeated user-testing sessions to iterate on comfort and durability with interchangeable tips.

EXPERIENCE

True North Bio-competition

Electrical Designer

2025 – Present

- Creating the electrical control framework for a lower-limb exoskeleton, utilizing EMG sensors to facilitate intuitive, signal-based movement for assistive applications.

BioTEC Conference

Sponsorship Lead

Dec 2025 – Present

- Securing partnerships and corporate outreach for Canada's largest student-run biotechnology conference.

Lifeguard & Swimming Instructor

Instructor, NLS, CPR Certified

Town of Milton

June 2023 – 2025

- Selected for highly competitive outdoor pool placement (Top 4% of 250+ staff) based on performance.

Biomechatronics Design Team

Mechanical Designer

2025 – Present

- Developing mechanical components for assistive orthotic devices, focusing on structural integrity.

EDUCATION EXPERIENCE

University of Waterloo

B.A.Sc. in Biomedical Engineering

Waterloo, ON

Class of 2030

Honors: Governor General's Academic Medal, Sir Isaac Newton Physics Award, TDCH Valedictorian, HHCS Valedictorian.

WEEF Representative

Waterloo Engineering Endowment Foundation

Sept 2025 – Present

- Evaluated technical proposals for student project funding and represented class interests to faculty.

Personal Tutor

Mathematics and Physics

Sept 2023 – June 2025

- Instructed high school students in calculus, physics, biology, and chemistry, translating complex concepts.